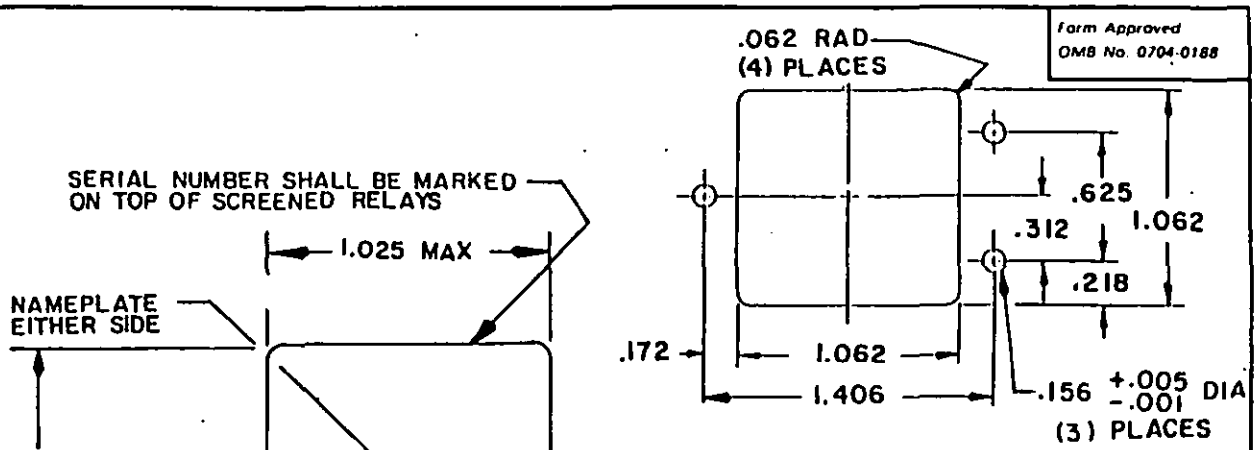
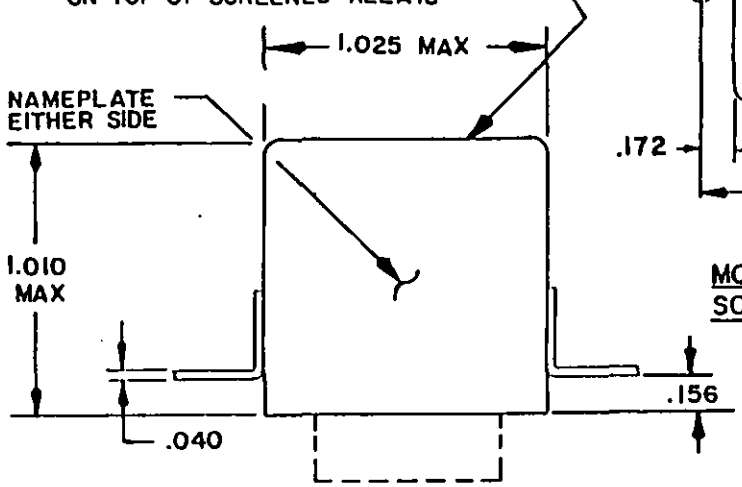
THIS SPECIFICATION IS APPROVED FOR USE BY the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(s) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-R-6106

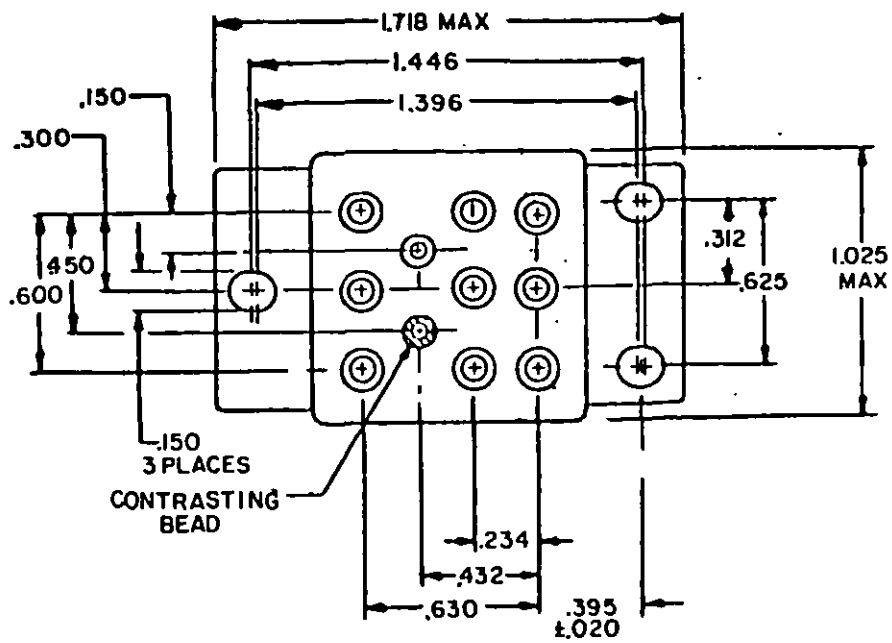
Form Approved
OMB No. 0704-0188



MOUNTING HOLE LAYOUT FOR SOLDER-HOOK TERMINALS



Inches	mm
.001	0.03
.005	0.13
.020	0.51
.040	1.02
.062	1.57
.150	3.81
.156	3.96
.172	4.37
.218	5.54
.234	5.94
.300	7.62
.312	7.96
.395	10.03
.432	10.97
.450	11.43
.600	15.24
.625	15.88
.630	16.00
1.010	25.65
1.025	26.04
1.062	26.97
1.396	35.46
1.406	35.71
1.446	36.73



TERMINAL LAYOUT FOR ALL SOLDER-HOOK, DC SOLDER-PIN AND SOCKET-PIN RELAYS

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).

FIGURE 1. Outline drawing (for details see tables I and II).

(J) Denotes Changes

PREPARING ACTIVITY: Air Force - 85 CUSTODIANS: ARMY - NAVY - AIR FORCE - 85 DLA - REVIEW: Air Force - 11, 99 USER: PROJECT NUMBER: 5945-F707	MILITARY SPECIFICATION SHEET TITLE RELAYS, 25 AMPERES, 3PDT, ALL WELDED, TYPE I, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE	SPECIFICATION SHEET NUMBER MS27743J(USAF) 18 JAN 1990 SUPERSEDING MS27743H(USAF) 3 Nov 1989 AMSC N/A FSC 5945

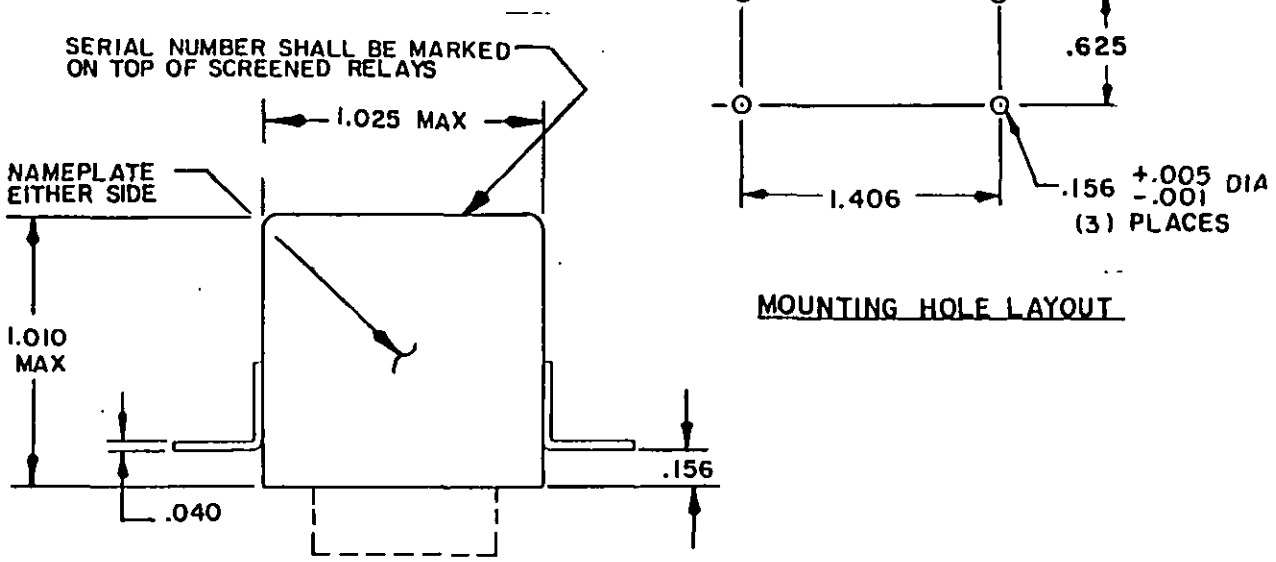
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil - Downloaded: 2026-10-29T10:50Z
Check the source to verify that this is the current version before use.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-R-6106

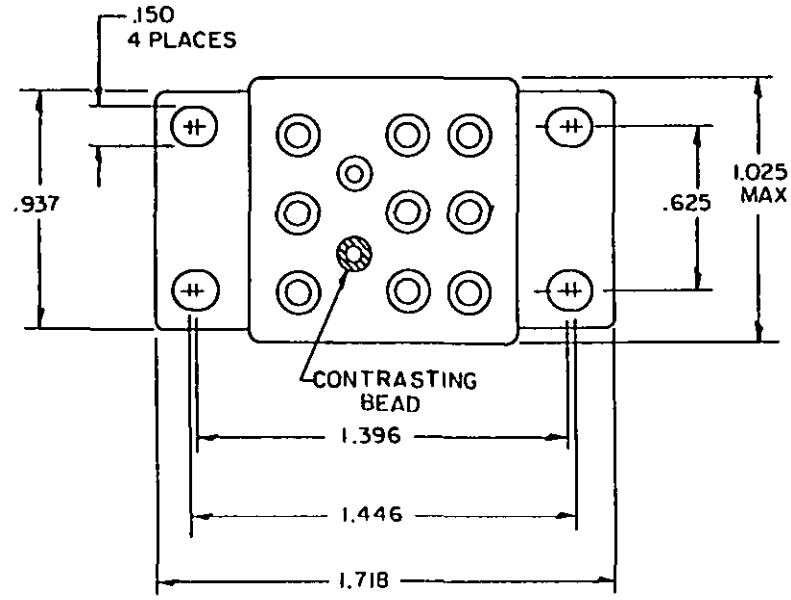
THIS SPECIFICATION IS APPROVED FOR USE BY the Department of the Force and is available for use by all Departments and Agencies the Department of Defense.

Form Approved
OMB No. 0704-0188



MOUNTING HOLE LAYOUT

Inches	mm
.001	0.03
.005	0.13
.040	1.02
.150	3.81
.156	3.96
.625	15.88
.937	23.80
1.010	25.65
1.025	26.04
1.396	35.46
1.406	35.71
1.446	36.73
1.718	43.64



TERMINAL LAYOUT FOR ALL A C SOCKET PINS (GOLD PLATED)

FIGURE 2. Outline drawing (for details see tables I and II).

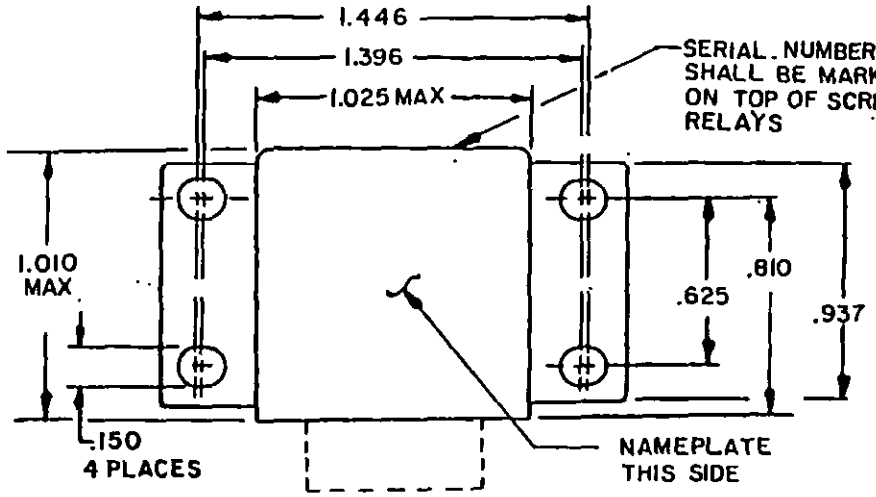
PREPARING ACTIVITY: Air Force - 85 CUSTODIANS: ARMY - NAVY - AIR FORCE - 85 DLA - REVIEW: Air Force - 11, 99 USER: PROJECT NUMBER: 5945-F707 DISTRIBUTION STATEMENT	MILITARY SPECIFICATION SHEET TITLE RELAYS, 25 AMPERES, 3PDT, ALL WELDED, TYPE I, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE	SPECIFICATION SHEET NUMBER MS27743J(USAF) SUPERSEDING MS27743H(USAF) 3 Nov 1989 AMSC N/A FSC 5945

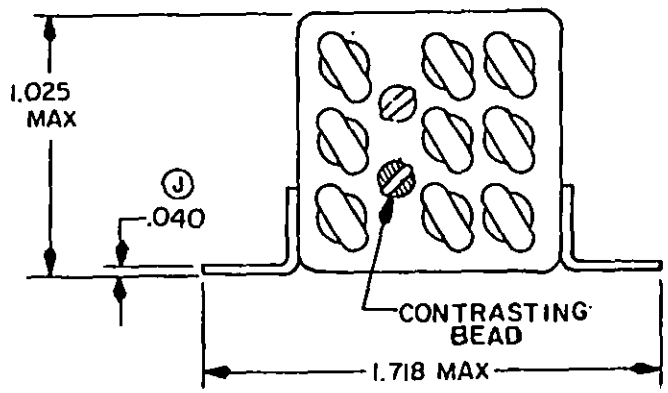
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:50Z
Check the source to verify that this is the current version before use.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(s) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-R-6106

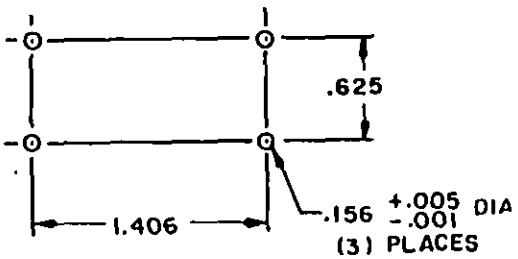
IS SPECIFICATION IS APPROVED FOR USE BY the Department of the Air Force d is available for use by all Departments and Agencies of the partment of Defense.

Form Approved
OMB No. 0704-0188

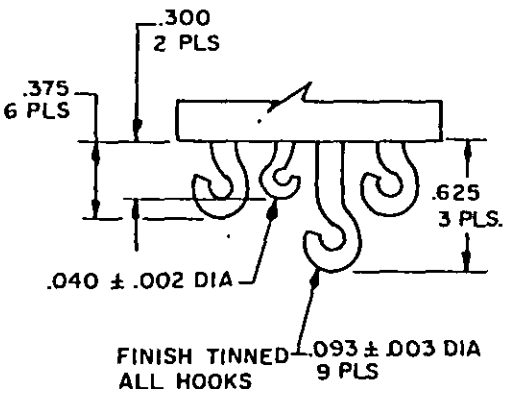




TERMINAL LAYOUT FOR ALL SOLDER-HOOK AND SOLDER-PIN (TIN PLATE FINISH) RELAYS



MOUNTING HOLE LAYOUT



SOLDER-HOOK TERMINAL

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).

FIGURE 3. Outline drawing (for details see tables I and II).

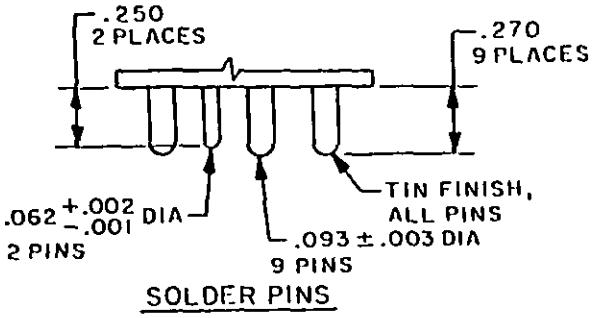
PREPARING ACTIVITY: Air Force - 85 CUSTODIANS: ARMY - NAVY - AIR FORCE - 85 DLA - REVIEW: Air Force - 11, 99 USER: PROJECT NUMBER: 5945-F307 DISTRIBUTION STATEMENT	MILITARY SPECIFICATION SHEET TITLE RELAYS, 25 AMPERES, 3PDT, ALL WELDED, TYPE I, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE	SPECIFICATION SHEET NUMBER MS27743J(USAF) SUPERSEDING MS27743H(USAF) AMSC N/A FSC 5945

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:50Z
Check the source to verify that this is the current version before use.

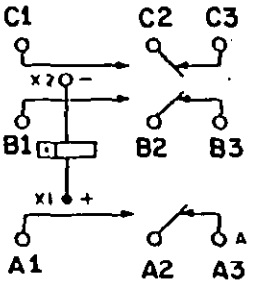
THIS SPECIFICATION IS APPROVED FOR USE BY the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(s) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-R-6106

Form Approved
OMB No. 0704-0188

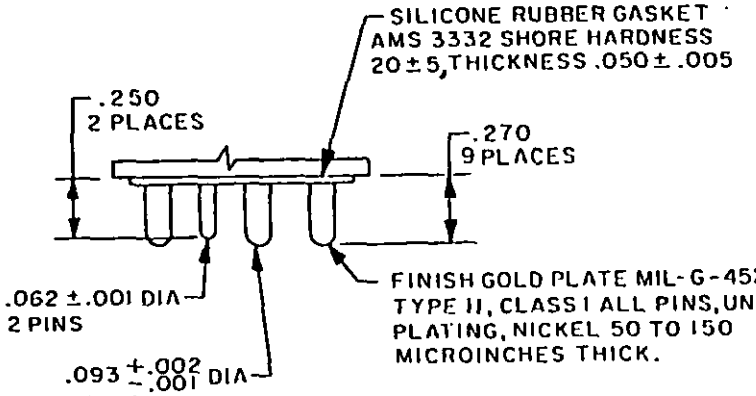


SOLDER PINS

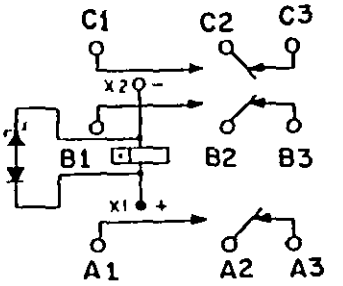


CIRCUIT A

(POLARIZED MONOSTABLE)
(SEE NOTES 7, 8, 9, AND 10)



SOCKET PINS



CIRCUIT B

(SEE NOTE 10)

Inches	mm	Inches	mm	Inches	mm
.001	0.03	.150	3.81	.937	23.80
.002	0.05	.156	3.96	1.010	25.65
.003	0.08	.250	6.35	1.025	26.04
.005	0.13	.270	0.97	1.396	35.46
.020	0.51	.300	11.43	1.406	35.71
.040	1.02	.375	15.24	1.446	36.73
.062	1.57	.625	15.88		
.093	2.36	.810	20.57		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (.25 mm).
4. Terminal numbers need not appear on the relay header provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. For design feature purposes, this standard takes precedence over acquisition documents referenced herein.
6. Referenced documents shall be of the issue in effect on date of invitation for bid.
7. Coil polarity not applicable to ac versions 7,8,9,13,14,15,19,20,21,25,26, and 27.
8. DC versions of this relay must not operate or be damaged by reverse polarity. Semiconductors shall not be used for this purpose.
9. Permanent magnet drive consists of permanent magnet with its flux path switched and combined with the electro-magnet flux.
10. For details see table I.

FIGURE 3. Outline drawing (for details see tables I and II) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:50Z
Check the source to verify that this is the current version before use.

THIS SPECIFICATION IS APPROVED FOR USE BY the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-R-6106

Form Approved
OMB No 0704-0188

TABLE I. Dash numbers and characteristics.

MS part number MS27743-	Type	Coil frequency (Hz) 1/ 2/	Coil 1/ Suppression (transient voltage, back EMF)	Terminal type 3/ (J)	Mounting	Mating socket	Circuit	Weight max Pounds	Weight max Grams
4 4/ 5/	I	dc	N/A	Solder hook	Figure 1	N/A	A	.18	82
5 4/ 5/	I	dc	N/A	Solder hook	Figure 3	N/A	A	"	"
6 4/ 5/	I	dc	N/A	Socket pin	Figure 1	MIL-S-12883/48	A	"	"
7 5/	I	400	N/A	Solder hook	Figure 1	N/A	A	"	"
8 5/	I	400	N/A	Solder hook	Figure 3	N/A	A	"	"
9 5/	I	400	N/A	Socket pin	Figure 2	MIL-S-12883/48	A	"	"
10 6/	I	dc	N/A	Solder hook	Figure 1	N/A	A	"	"
11 "	I	dc	N/A	Solder hook	Figure 3	N/A	A	"	"
12 "	I	dc	N/A	Socket pin	Figure 1	MIL-S-12883/48	A	"	"
13 "	I	400	N/A	Solder hook	Figure 1	N/A	A	"	"
14 "	I	400	N/A	Solder hook	Figure 3	N/A	A	"	"
15 6/	I	400	N/A	Socket pin	Figure 2	MIL-S-12883/48	A	"	"
16 5/	I	dc	42 V dc max	Solder hook	Figure 1	N/A	B	"	"
17 "	I	dc	42 V dc max	Solder hook	Figure 3	N/A	B	"	"
18 "	I	dc	42 V dc max	Socket pin	Figure 1	MIL-S-12883/48	B	"	"
19 "	I	50/400	N/A	Solder hook	Figure 1	N/A	A	"	"
20 "	I	50/400	N/A	Solder hook	Figure 3	N/A	A	"	"
21 5/	I	50/400	N/A	Socket pin	Figure 2	MIL-S-12883/48	A	"	"
22 6/	I	dc	42 V dc max	Solder hook	Figure 1	N/A	B	"	"
23 "	I	dc	42 V dc max	Solder hook	Figure 3	N/A	B	"	"
24 "	I	dc	42 V dc max	Socket pin	Figure 1	MIL-S-12883/48	B	"	"
25 "	I	50/400	N/A	Solder hook	Figure 1	N/A	A	"	"
26 "	I	50/400	N/A	Solder hook	Figure 3	N/A	A	"	"
27 6/	I	50/400	N/A	Socket pin	Figure 2	MIL-S-12883/48	A	"	"
28 5/	I	dc	N/A	Solder pin	Figure 1	N/A	A	"	"
29 5/	I	dc	N/A	Solder pin	Figure 3	N/A	A	"	"
30 6/	I	dc	N/A	Solder pin	Figure 1	N/A	A	"	"
31 6/	I	dc	N/A	Solder pin	Figure 3	N/A	A	"	"
32 5/	I	dc	42 V dc max	Solder pin	Figure 1	N/A	B	"	"
33 5/	I	dc	42 V dc max	Solder pin	Figure 3	N/A	B	"	"
34 6/	I	dc	42 V dc max	Solder pin	Figure 1	N/A	B	"	"
35 6/	I	dc	42 V dc max	Solder pin	Figure 3	N/A	B	"	"

NOTES:

1/ JAN TX or equivalent screened semiconductors shall be used in screened relays with internal coil suppression or ac coil ratings after 30 December 1980. Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode.

2/ Diodes shall have a peak inverse voltage of 600 V dc minimum when used.

3/ All solder pins shall be tin plated. All socket pins shall be gold plated in accordance with MIL-G-45204, type II, class 1.

4/ MS27743-1 of the previous edition is cancelled and superseded by MS27743-4, effective 1 August 1980.
MS27743-2 of the previous edition is cancelled and superseded by MS27743-6, effective 1 August 1980.
MS27743-3 of the previous edition is cancelled and superseded by MS27743-5, effective 1 August 1980.

5/ For Government logistic support, see table V.

6/ Screened relays shall be tested to the group A ER requirements only and additional life testing is not required. For group A inspection, operational reliability shall be performed at the alternate low level run in.

PREPARING ACTIVITY: Air Force - 85 CUSTODIANS: ARMY - NAVY - AIR FORCE 85 DLA - REVIEW: Air Force - 11, 99 USER: PROJECT NUMBER: 5945-F707	MILITARY SPECIFICATION SHEET TITLE RELAYS, 25 AMPERES, 3 PDT, ALL WELDED, TYPE I, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE	SPECIFICATION SHEET NUMBER MS27743J(USAF) SUPERSEDING MS27743H(USAF) AMSC N/A FSC 5945

DISTRIBUTION STATEMENT A Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil - Downloaded: 2026-10-23T16:50Z
Check the source to verify that this is the current version before use.

HIS SPECIFICATION IS APPROVED FOR USE BY the Department of the
\ir Force and is available for use by all Departments and Agencies
)f the Department of Defense.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(s) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-R-6106

PREPARING ACTIVITY: Air Force - 85
CUSTODIANS: ARMY - NAVY - AIR FORCE - 85 - DLA -
REVIEW: Air Force - 11, 99
USER:
PROJECT NUMBER: 5945-F707
DISTRIBUTION STATEMENT

MILITARY SPECIFICATION SHEET
TITLE RELAYS, 25 AMPERES, 3PDT, ALL WELDED, TYPE I, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE

SPECIFICATION SHEET NUMBER
MS27743J(USAF)
SUPERSEDING
MS27743H(USAF)
AMSC N/A
FSC 5945

TABLE II. Operating characteristics.

MS part no. 1/	Coil data												Time (milliseconds - maximum)						
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 2/	Drop out voltage 2/		Operate 3/	Release 4/	Contact bounce				Break bounce NO only ⓙ
		Volts 1/	Freq Hz	Res Ω ±10% at +25°C	Volts	Amp	Normal test 2/	High temp test	Cont current test						Main		Aux		
															NO	NC	NO	NC	
4, 5, 6, 10, 11, 12, 28, 29, 30, 31	X1,X2	28	dc	290	29	0.12	18	19.8	22.5	7.0	1.5		15	15	1	1	--	--	
7, 8, 9, 13, 14, 15	X1,X2	115	400	---	122	0.04	90	95.4	103.5	30	5.0		20	50	1	1	--	--	0.1
16, 17, 18, 22, 23, 24, 32, 33, 34, 35	X1,X2	28	dc	290	29	0.12	18	19.8	22.5	7.0	1.5		15	15	1	1	--	--	0.1
19, 20, 21, 25, 26, 27	X1,X2	115	50/ 400	---	122	0.03	95	100	105	40	5		20	50	1	1	--	--	0.1

NOTES:
1/ Caution: The use of any coil voltage less than the nominal coil voltage will compromise the operation of the relay.
2/ Over temperature range
3/ With nominal coil voltage.
4/ From nominal coil voltage.

Form Approved
OMB No 0704-0188

Source: https://assist.dla.mil -- Downloaded: 2020-10-23T16:50Z
Check the source to verify that this is the current version before use.

HIS SPECIFICATION IS APPROVED FOR USE BY the Department of the
Air Force and is available for use by all Departments and Agencies
of the Department of Defense.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(s) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-R-6106

PREPARING ACTIVITY: Air Force - 85
CUSTODIANS: ARMY - NAVY - AIR FORCE - 85 DLA -
REVIEW: Air Force - 11, 99
USER:
PROJECT NUMBER: 5945-F707
DISTRIBUTION STATEMENT

MILITARY SPECIFICATION SHEET
TITLE RELAYS, 25 AMPERES, 3PDT, ALL WELDED, TYPE 1, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE

SPECIFICATION SHEET NUMBER: MS27743J(USAF)
SUPERSEDING: MS27743H(USAF)
AMSC N/A
FSC 5945

TABLE III. Rated contact load (amperes per pole), case grounded. 1/

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1-phase				115/200 V ac, 3-phase				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive 2/	50	25	25	--	--	25	--	--	--	25	--	--	--	
Inductive	10	12	12	--	--	--	--	--	--	--	--	--	--	
Inductive	20	--	--	--	--	15	--	--	--	15	--	--	--	
Motor	50	10	10	--	--	10	--	--	--	10	--	--	--	
Lamp	50	5	5	--	--	5	--	--	--	5	--	--	--	
Transfer, load	--	--	--	--	--	--	--	--	--	--	--	--	--	3/
Mechanical life reduced current	200	6	6	--	--	6	--	--	--	6	--	--	--	
Intermediate current	Applicable per specification													

1/ Absence of values indicates that relay is not rated for 3-phase application.
2/ For full rated load, maximum temperature and altitude use number 12 wire or larger. Relays shall be mounted to limit mounting bracket temperature to +160°C maximum.
3/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplied at rating indicated.

Form Approved
OMB No 0704-0188

Source: https://assist.dla.mil -- Downloaded: 2020-10-23T16:50Z
Check the source to verify that this is the current version before use.

HIS SPECIFICATION IS APPROVED FOR USE BY the Department of the
Air Force and is available for use by all Departments and Agencies
of the Department of Defense.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(s) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-R-6106

PREPARING ACTIVITY: Air Force - 85
CUSTODIANS: ARMY - NAVY - AIR FORCE - 85 DLA -
REVIEW: Air Force - 11, 99
USER:
PROJECT NUMBER: 5945-F707
DISTRIBUTION STATEMENT

MILITARY SPECIFICATION SHEET
TITLE RELAYS, 25 AMPERES, 3PDT, ALL WELDED, TYPE I, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE

SPECIFICATION SHEET NUMBER MS27743J(USAF)
SUPERSEDING MS27743H(USAF)
AMSC N/A
FSC 5945

Environmental characteristics
(Applicable to figures 1 and 2, see table I.)

Temperature range	-70°C to +125°C
Maximum altitude rating	300,000 ft
Shock G-level	200 G
Duration	6 ms
Max duration contact opening	10 µs
Vibration - sinusoidal	
G-level	30 G
Frequency range	10-3000 Hz
Vibration-random	
Applicable specification	MIL-STD-202
Method	214
Test condition	IG 0.4 G^2/Hz
Duration	15 minutes each plane
Acceleration	15 G

Electrical characteristics
Insulation resistance, initial 100 megohms
After life or environmental tests 50 megohms
Dielectric strength (sea level)

	Inital	After life tests
Coil to case	1000 V rms	1000 V rms
Aux contacts	---	---
All other points	1250 V rms	1000 V rms

Dielectric strength (sea level) 1/

	80,000 ft	300,000 ft
Coil to case	350 V rms	500 V rms
Aux contacts	---	---
All other points	350 V rms	500 V rms

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	50 amperes dc, 80 amperes ac
Rupture current	60 amperes dc, 100 amperes ac
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

1/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.

Electrical characteristics
(Applicable to figure 3, see table I.)

	-70°C to +125°C
	300,000 ft
	100 G
	6 ms
	10 µs
	20 G
	10-3000 Hz
	MIL-STD-202
	214
	IE 0.2 G^2/Hz
	15 minutes each plane
	15 G

TABLE IV. Time-current relay characteristics at +25°C. 1/ 2/

1	30 A - hour
2	50 A - 5.0 seconds
3	100 A - 1.2 seconds
4	250 A - 0.2 second
5	350 A - 0.1 second

1/ Caution: Compare with time-current characteristics of the associated circuit protective device.

2/ Time-current relay characteristics at +25°C. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of five different current levels and 115/200 V ac 400 Hz 3-phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normaly closed contacts of each relay. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

Form Approved OMB No 0704-0188

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:30Z
Check the source to verify that this is the current version before use.

THIS SPECIFICATION IS APPROVED FOR USE BY the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(s) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-R-6106

Form Approved
OMB No. 0704-0188

Cross reference for Government support: See table V.

TABLE V. Cross reference for Government logistical support.

MS Part or Identifying Number MS27743-	Support with MS Part or Identifying Number MS27743-
4	10
5	11
6	12
7	13
8	14
9	15
16	22
17	23
18	24
19	25
20	26
21	27
28	30
29	31
32	34
33	35

PREPARING ACTIVITY: Air Force - 85
CUSTODIANS: ARMY - NAVY - AIR FORCE - 85 DLA -
REVIEW: Air Force - 11, 99
USER:
PROJECT NUMBER: 5945-F707
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

MILITARY SPECIFICATION SHEET
TITLE RELAYS, 25 AMPERES, 3PDT, ALL WELDED, TYPE I, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE

SPECIFICATION SHEET NUMBER
MS27743J(USAF)
SUPERSEDING
MS27743H(USAF)
AMSC N/A FSC 5945

Source: https://assist.dla.mil - Downloaded: 2020-10-29T10:50Z
Check the source to verify that this is the current version before use.